FINANCE · ETF COURSE · LECTURE 1

Exchange Traded Funds

Study notes + self-test MCQs · BCG Global Asset Management Report · Larry Fink Letters 2025 & 2026 · Answers at the end

CONTENTS

⚡ EXAM REMINDER

Exam = last 1.5 hours of class. MCQs follow each section — answers revealed on the final pages only.

SECTION 01

The Asset Management Industry

$128T
TOTAL AUM TODAY

$36T
AUM 20 YEARS AGO

7%
ANNUAL GROWTH RATE

$20T
ETF MARKET SIZE

Key insight: Most 7%/year growth comes from market appreciation, not new money. Net flows are only 1–3%/year. MSCI ACWI IMI ≈ $100T (all listed stocks). Bloomberg Global Agg ≈ $70T (all govt & corporate debt).

THE 5 SEGMENTS OF ASSET MANAGEMENT

1 **Passives**
$30T AuM (23% of total) — only
5% of industry profitability
. Includes ETFs (50% of passives), index funds, and mandates.

2 **Active Specialties**
Niche active strategies targeting specific sectors or styles.

3 **Solutions, LDI & Multi-Assets**
Liability-Driven Investing — complex non-linear payoffs. Combines equities, fixed income, etc.

4 **Active Core**
Promises alpha (outperformance) vs index. Higher fees. Squeezed from both sides by passives and alternatives. Passive + Alternatives market share: was 20%, now 41%.

5 **Alternatives**
Hedge funds, private debt, commodities, infrastructure. 18% AuM but
58% of profitability
. Most profitable segment. Some private debt funds have closed due to redemption pressure.

PROFITABILITY & CHALLENGES

SEGMENT	AUM SHARE	PROFIT SHARE	TREND
Passives	23%	5%	Growing fast, low margin
Active Core	~40%	Shrinking	Squeezed from both extremes
Alternatives	18%	58%	Most profitable; growing

Cost-income ratio: Industry average ~70%. Amundi ~50%. 70–80% of net flows going into passives → fee compression across the industry.

MCQ — SECTION 01 · 5 QUESTIONS

The Asset Management Industry

Q1. What is the approximate total size of global AuM today?

- A $36 trillion
- B $70 trillion
- C $128 trillion
- D $200 trillion

Q2. Although passives hold 23% of AuM, what share of total industry profitability do they represent?

- A 23%
- B 5%
- C 18%
- D 58%

Q3. Alternatives represent 18% of AuM. What share of profitability do they capture?

- A 18%
- B 30%
- C 41%
- D 58%

Q4. The combined market share of passives + alternatives has risen from 20% to what figure?

- A 25%
- B 33%
- C 41%
- D 50%

Q5. What percentage of net flows now goes into passive products?

- A 30–40%
- B 50–60%
- C 70–80%
- D 90–100%

SECTION 02

Why Passive Investing Is Winning

10%
US ACTIVE BEAT INDEX (15Y)

3%
EU ACTIVE BEAT INDEX (10Y)

30–50%
BEAT INDEX OVER 1 YEAR

13%
LONG-RUN AVERAGE

Core rule: The longer the period, the worse active managers perform. 1 year: 30–50% beat. 10+ years: only 10–20%, and declining.

WHEN ACTIVE CAN WIN (MORNINGSTAR)

⚠ ACTIVE UNDERPERFORMS — BUY ETF

→ French equities: only 5% chance
→ US large-cap equities
→ Most developed market equities
→ Fully-researched large caps

✅ ACTIVE HAS EDGE

→ India & Switzerland (~50% beat)
→ Small caps (less analyst coverage)
→ Emerging markets (info asymmetry)
→ Post-COVID fixed income (rising rates)

Why small caps & EM? Large caps have hundreds of research papers — prices are fully efficient. Small caps may have zero analyst coverage → price inefficiencies remain exploitable.

Fixed income post-COVID: Over 50% of active bond managers beat ETFs. Rates rose from ~0% → differentiation opportunities now exist that didn't before.

ACTIVE SHARE

Formula: $\Sigma|\text{holding weight} - \text{index weight}| \div 2$. Range: 0% (= index) → 100% (completely different). Bigger funds → smaller active share → harder to generate alpha. "Closet index funds" = behave like an index but charge active fees.

MCQ — SECTION 02 · 5 QUESTIONS

Why Passive Investing Is Winning

Q1. Per SPIVA, what % of European active managers beat their index over 10 years?

- A 10%
- B 15%
- C 30%
- D 3%

Q2. If a fund has the same portfolio as its benchmark, what is its Active Share?

- A 0%
- B 50%
- C 100%
- D Cannot be calculated

Q3. Why have active fixed income managers outperformed more since COVID?

- A Bond markets became less liquid
- B Rising rates created more differentiation opportunities
- C ETF fees increased sharply
- D Govt bonds were excluded from indices

Q4. In which market are you most likely to find alpha through active management?

- A US large-cap equities
- B French equities
- C Indian equities
- D UK large-cap equities

Q5. What is a "closet index fund"?

- A A fund that only invests in real estate
- B A passive fund that secretly buys active positions
- C A fund so large it cannot deviate from the index, yet charges active fees
- D A fund with a 0% expense ratio

SECTION 03

Key Players: BlackRock & Vanguard

🏦 BLACKROCK

→ Founded 1988 by Larry Fink (CEO)
→ Manages >$11.6 trillion — #1 globally
→ 2/3 of funds are passive
→ Bought BGI (iShares) 2010 for $13.5bn → #1 ETF
→ Sells Aladdin software to all PMs ($1.6bn+ revenue)
→ First to launch a Bitcoin spot ETF (2024)
→ Larry Fink's annual letter = must-read
→ Acquiring HPS (private credit), GIP (infrastructure), Preqin (data)
→ Dominant in Europe: 42% ETF market share

🏛 VANGUARD

→ Founded by John C. Bogle
→ 1st index fund ever: Vanguard 500 (1975)
→ Private — fund holders = shareholders
→ "No man can serve two masters" (fiduciary)
→ Expense ratios avg. <10bps
→ Bigger they get → lower their fees
→ 20M clients, mostly US direct-to-consumer
→ Only 1 employee in France
→ VOO has now overtaken SPY in AuM

The Vanguard Effect: US outflows from active funds almost perfectly mirror inflows into ETFs. Active avg fee = 71.4bps vs passive 13.5bps — passive is 5× cheaper.

Two growth strategies all major asset managers pursue: (1) ETFs and (2) Alternatives.

MCQ — SECTION 03 · 4 QUESTIONS

Key Players

Q1. How did BlackRock become the world's #1 ETF firm?

- A They invented the first index fund in 1975
- B They launched the first Bitcoin ETF
- C They acquired BGI (iShares) for $13.5bn in 2010
- D They merged with Vanguard

Q2. Why is Vanguard structured as a private company owned by its fund holders?

- A To avoid corporate taxes
- B To serve only one master — focus entirely on fund holders
- C Required by SEC regulation
- D To allow unlimited illiquid investment

Q3. What was Bogle's key insight when creating the first index fund?

- A Algorithmic trading would replace fund managers
- B Most mutual funds don't outperform index funds after fees
- C Bonds were better than equities long-term
- D ETFs would replace stocks

Q4. What are the two growth strategies all major asset managers now pursue?

- A Fixed income and technology
- B Active core and solutions
- C ETFs and Alternatives
- D Emerging markets and real estate

SECTION 04

What Is an ETF?

Core definition: A UCITS fund listed on an exchange, tradeable intraday via bid/ask quotation — like a stock. 90% of European ETFs are index-based.

UCITS VS AIF

FEATURE	UCITS	AIF (ALTERNATIVE)
For whom	Retail investors	Professionals / UHNW (>€100k min)
Liquidity	Daily, intraday trading	Illiquid, complex
EU passport	✓ 26 EU countries	Limited

UCITS INVESTOR PROTECTIONS

5/10/40 Rule
No single holding above 10%. All positions 5–10% must sum to <40%. Max leverage = 2×. Index-specific rule: 20/35.

Segregated Accounts
All assets held at a **separate custodian**. Protects against PM fraud (cf. Madoff scandal — impossible under UCITS).

Counterparty Risk
Capped at 10% for derivatives. In practice usually <5%.

3 Key Documents
Prospectus (full detail), **KID** (3 pages: fees, track record, strategy), Annual **Audit** by independent firm (Deloitte, EY, etc.).

KEY ETF METRICS — MUST KNOW

TER
Total Expense Ratio — annual fee ~25bps in Europe. Does NOT include trading spread.

Spread
Buy vs sell price difference. Target <10bps. SPY = 0.5bps.

Tracking Diff (TD)
ETF return − index return. Should be ≈0. Hurt by fees; helped by securities lending income. = Pure performance measure.

Tracking Error (TE)
Annualised volatility of TD. Measures consistency. AMF limit: TE <1% (5% for volatile indices).

NAV / iNAV
NAV = end-of-day value. iNAV = real-time indicative value, calculated every ~15 seconds. Market price stays near iNAV via arbitrage.

PCF
Portfolio Composition File — published by ETF issuer before 9am daily. Market makers use it to price the ETF.

SPY — First ETF (1993), ~$500bn AuM, spread = 0.5bps, $25bn daily volume. Now second to Vanguard's VOO.

MCQ — SECTION 04 · 5 QUESTIONS

What Is an ETF?

Q1. What is the key difference between Tracking Difference and Tracking Error?

- A They are the same thing
- B TD = pure performance gap; TE = volatility (consistency) of that gap over time
- C TD measures fees; TE measures liquidity
- D TD is used in Europe; TE in the US

Q2. What is the AMF's maximum Tracking Error to classify a fund as an index fund?

- A 0.1%
- B 1% (5% for highly volatile indices)
- C 5% for all indices
- D 10%

Q3. UCITS 5/10/40 rule: all positions between 5% and 10% must sum to less than:

- A 10%
- B 20%
- C 30%
- D 40%

Q4. What is the PCF, and when must it be published?

- A Price Correction File — at market close
- B Portfolio Composition File — before 9am daily by the ETF issuer
- C Premium/Cost Formula — weekly
- D Performance Comparison Figure — monthly

Q5. Which of the following can positively impact an ETF's Tracking Difference?

- A High management fees
- B Securities lending income
- C Wide bid-ask spreads
- D High tracking error

SECTION 05

ETF Growth & Market Structure

18%
ETF CAGR (10 YEARS)

15k+
ETFS GLOBALLY

$3T
EUROPEAN ETF AUM

42%
ISHARES EU MARKET SHARE

Scale: ETF industry grew 18%/year vs 7% for all asset management. ETFs = $20T / $128T total = ~20% of all AuM. There are now more ETFs (15k+) than listed equities (MSCI IMI = 8,600).

4 SECULAR GROWTH DRIVERS

1. **The Wealth Wave**
Rebate ban on advisor kickbacks (UK, US, Netherlands done; France likely soon). Active fund kickbacks = 1–2%/year vs 25bps for ETFs (5×). When banned, advisors choose cheapest products → ETFs.

2. **The Retail Wave**
More retail investors discovering low-cost products via apps. Germany = biggest European retail ETF market (10M+). Pension caps incentivise self-investment.

3. **The Green Wave**
ESG innovation is ideal for ETFs (rule-based, data-driven). Strong institutional demand for climate-aligned indices in Europe.

4. **The International Wave**
UCITS ETFs are globally accessible. Key tax advantage: UCITS funds are tax-protected until you sell. US funds are tax-transparent (gains taxed annually). Many Asian investors use UCITS ETFs for this reason.

MCQ — SECTION 05 · 4 QUESTIONS

ETF Growth & Market Structure

Q1. What is the key tax advantage of UCITS ETFs for international investors?

- A They pay 0% tax on all gains forever
- B Gains are tax-protected until sale — unlike US funds taxed annually
- C Dividends automatically reinvested tax-free
- D Exempt from capital gains in all jurisdictions

Q2. What is the primary mechanism of the "Wealth Wave" driving ETF adoption?

- A Increasing number of billionaires in ETFs
- B Younger investors preferring digital products
- C Banning of advisor rebates forces recommendation of lowest-cost products
- D Central banks buying ETFs in QE

Q3. What share of the European ETF market does iShares hold?

- A 20%
- B 30%
- C 42%
- D 65%

Q4. How much did European ETFs collect in net flows last year?

- A $100bn
- B $259bn
- C $333bn
- D $500bn

SECTION 06

Retail Evolution in Europe & PEA

1M	FRENCH ETF BUYERS (2026)
45%	FRENCH INVESTORS UNDER 35
10M	EU MONTHLY ETF SAVERS
7M	FRENCH WITH A PEA

THE FRENCH PEA (PLAN ÉPARGNE ACTION)

Tax benefit
After 5 years: no income tax — only social tax (~15.5% vs 30% flat tax).

Limit & scope
Max €150k. At least 75% must be in European companies. PEA-PME sibling for small caps.

PEA trick
MSCI World/S&P 500 ETFs qualify via **synthetic replication** — substitute basket = European equities → satisfies the 75% EU rule while giving global exposure.

WPEA ETF
iShares MSCI World Swap PEA UCITS ETF: €20–70M/month; 7bps spread; €400M in 10 months.

Savings plans: 10M Europeans invest monthly, no brokerage fees. Fee war: Amundi = 25bps, Boursorama = 20bps.

MCQ — SECTION 06 · 4 QUESTIONS

Retail Evolution & PEA

Q1. How long must you hold assets in a PEA to pay only social tax on gains?

- A 1 year
- B 3 years
- C 5 years
- D 10 years

Q2. How can an MSCI World ETF (60% US) qualify for a PEA (requires 75% EU assets)?

- A The fund holds only European stocks
- B Synthetic replication: substitute basket = European equities
- C The AMF grants a waiver for large ETFs
- D It cannot qualify

Q3. What is the maximum amount that can be invested in a PEA?

- A €50,000
- B €100,000
- C €150,000
- D €250,000

Q4. What proportion of active French ETF investors are under 35?

- A 20%
- B 30%
- C 45%
- D 60%

SECTION 07

The Index Industry

Definition: A rule-based algorithm publishing one value per day representing an asset class, sector, or strategy. No PM discretion. Can be: long only, short, leveraged, covered call, protective put, etc.

DIVIDEND TREATMENT — 3 VERSIONS

VERSION	TREATMENT	EXAMPLE
Price Return (PR)	Dividends NOT reinvested. Stock at 100, pays 2€ → drops to 98.	CAC 40, S&P 500 (quoted)
Gross Total Return (GTR)	Dividends reinvested BEFORE tax (full 2€).	DAX (unusual — uses GTR)
Net Total Return (NTR)	Dividends reinvested AFTER max withholding tax. If tax = 50%, only 1€ reinvested.	Typical PM benchmark

Critical: Newspapers always show Price Return. DAX = total return (comparisons misleading). NTR is the easiest benchmark to beat — uses maximum possible tax rate. European dividend yield ≈ 3–4%/year = gap between PR and TR.

WEIGHTING METHODS

Market Cap
Weight = market value. Theoretically optimal (Markowitz). Risk: mega-caps dominate.

Equal Weight
All same weight. Requires rebalancing. Good diversification but risky (equal weight on illiquid micro-caps).

Price Weighted
Weight = stock price. Nikkei uses this (historical habit).

Volume Weighted
Weight = trading volume → maximises liquidity.

MSCI ACWI IMI: 47 markets, 99% of companies. ~2/3 US equities. France = 2%, all Eurozone = 8%. China heavily discounted vs GDP (currency controls, emerging market status). "The only free thing in finance is diversification" — Markowitz.

MCQ — SECTION 07 · 5 QUESTIONS

The Index Industry

Q1. A stock is at 100 and pays a 3€ dividend. Its value in a Price Return index after the dividend?

- A 100 (unchanged)
- B 97
- C 103
- D 100 minus taxes

Q2. Which index is unusual because it uses a Total Return methodology?

- A CAC 40
- B S&P 500
- C DAX
- D Nikkei

Q3. Why is Net Total Return (NTR) considered the "easiest benchmark to beat"?

- A It includes leverage, depressing returns
- B It applies the maximum possible withholding tax on dividends
- C It excludes small-cap stocks
- D It uses Price Return methodology

Q4. Which weighting method does the Nikkei use?

- A Market cap weighted
- B Equally weighted
- C Price weighted
- D Volume weighted

Q5. Why is China under-represented in the MSCI ACWI IMI vs its GDP weight?

- A China is not in the index at all
- B Chinese companies are too small to qualify
- C Classified as emerging market, currency not freely convertible, limited access → large discount
- D MSCI excludes state-owned enterprises

SECTION 08

ESG & Climate Indices

EU CLIMATE BENCHMARK LABELS

LABEL	DAY-1 CO_2 CUT	EXCLUSIONS	ANNUAL TRAJECTORY
EU CTB (Climate Transition)	−30% vs parent	Tobacco + controversials	−7%/year
EU PAB (Paris Aligned)	−50% vs parent	Tobacco + controversials	−7%/year

Paris Agreement: <1.5°C warming. Net zero by 2050. Cut emissions 45% by 2030 (vs 2010) = −7%/year. SBTs = companies with auditor-approved emission plans → overweighted.

SCOPE DEFINITIONS

Scope 1
Direct emissions from company's own operations.

Scope 2
Indirect emissions from purchased electricity and energy.

Scope 3
All indirect emissions throughout the entire value chain (suppliers, customers, product use). The largest and hardest to measure.

ESG INTEGRATION SPECTRUM (MSCI)

ACWI → **ESG Screened** (basic exclusions) → **ESG Universal/Tilt** (keep 50%, adjust weights) → **ESG Leaders/Focus** (top sector performers) → **SRI** (top 25% per sector, excludes ~75% of universe)

MCQ — SECTION 08 · 4 QUESTIONS

ESG & Climate Indices

Q1. What is the required day-1 carbon emission reduction for an EU PAB?

- A −7%
- B −20%
- C −30%
- D −50%

Q2. What annual decarbonisation trajectory is required for BOTH EU CTB and EU PAB?

- A −3%/year
- B −5%/year
- C −7%/year
- D −10%/year

Q3. Scope 3 emissions refer to:

- A Direct emissions from company operations
- B Emissions from purchased electricity
- C Indirect emissions throughout the entire value chain
- D Competitor company emissions

Q4. Which ESG approach keeps only the top 25% of ESG performers per sector?

- A ESG Leaders
- B ESG Screened
- C ESG Universal
- D SRI

SECTION 08B

ESG Deep Dive: EU Framework & Climate Methods

From slides 92–96 of the course (Arnaud Llinas)

EU'S ACTION PLAN — THE 4 REGULATORY PILLARS

The EU is not just creating products — it is building an entire regulatory framework to redirect capital toward sustainable activities. Four pillars work together:

Taxonomy
An official EU classification system that defines what economic activities count as "green." Like a green label: companies and projects must prove they qualify. 6 environmental objectives including climate change mitigation, biodiversity, circular economy, water protection.

Disclosure
Fund managers and companies must be transparent about their climate risks and ESG practices. You can't just claim to be green — you have to prove it (SFDR regulation: Article 6, 8, 9 funds).

Benchmarks
The EU created two official climate benchmark labels — CTB and PAB — so investors know exactly what they're buying. All benchmark providers must also disclose alignment with EU emission targets for ALL their indices.

MiFID II & IDD
Distribution regulations now require financial advisors to ask clients about their ESG preferences before recommending any product. Sustainability preferences are now a mandatory part of the advice process.

> **Key insight:** The EU is using regulation across the entire investment chain — from how activities are classified, to how they are disclosed, benchmarked, and sold. This is why ESG adoption in Europe is so far ahead of the US.

FROM RISK TO IMPACT — A CONCEPTUAL SHIFT

OLD LOW CARBON BENCHMARKS	NEW EU CLIMATE BENCHMARKS (CTB/PAB)
Focused on risk — how exposed am I to carbon?	Focused on **impact** — am I contributing to emission reduction?
Static & backward-looking — what was carbon intensity last year?	Dynamic & forward-looking — what is the decarbonisation trajectory?
Relative decarbonisation — just be better than the benchmark	Absolute decarbonisation — follow the actual 1.5°C science

> **The shift in one sentence:** Old approach = protect my portfolio from climate risk. New approach = use my portfolio to drive real-world decarbonisation.

THREE METHODS OF CLIMATE INTEGRATION (MSCI)

① Reduce Emissions
MSCI Low Carbon Target Indexes. Minimise carbon intensity while controlling tracking error. Goal: same portfolio, less carbon. Most pragmatic approach. Example: CalSTRS (major US pension fund) allocated 20% of public equity to a low-carbon index to reduce emissions without big bets away from benchmark.

② Drive the Transition
MSCI Climate Action Indexes. Include the 50% of companies best prepared for the climate transition relative to sector peers. Logic: back the winners of the transition, not just avoid the losers. A dirty company actively investing in renewables might stay in. Rewards effort and direction. Example: Ilmarinen (Finnish pension fund).

③ Align
MSCI Climate Paris Aligned Indexes. Align with the 1.5°C Paris Agreement. Combines: minimise transition risk, minimise physical risk (floods/droughts), increase exposure to climate solutions, follow −7%/year decarbonisation trajectory. Most ambitious. Example: NZ Super Fund shifted 40% of portfolio to Paris-aligned indices.

CTB VS PAB — FULL DETAILED REQUIREMENTS

REQUIREMENT	EU CTB	EU PAB
Carbon intensity cut (day one) — Scope 1+2 (+Scope 3 for Oil & Gas/Mining; Scope 3 for all sectors within 4-year phase-in)	−30%	−50%
Baseline exclusions (both)	Controversial weapons, societal norms violators (UN Global Compact,	

	OECD guidelines), EU Taxonomy DNSH principle, Tobacco	
Fossil-related activity exclusions	None — can still hold coal, oil, gas	Strict: excludes coal (1%+ revenues), oil (10%+), natural gas (50%+), electricity above 100gCO_2/kWh
Decarbonisation trajectory	Minimum −7% per annum for both	
Exposure to High Climate Impact Sectors	Must maintain at least equal exposure to parent index — cannot just exclude all energy companies	
Bonus for SBTs	Overweight companies with Science-Based Targets approved by an auditor	

Key difference: CTB = no fossil exclusions, just less carbon overall. PAB = hard exclusions on coal, oil, and high-carbon electricity. PAB is dramatically stricter on what it will hold.

THE ACTIVE SHARE PROBLEM — WHEN MARKETS DON'T DECARBONISE FAST ENOUGH

Both CTB and PAB must follow −7%/year. But what if the **real market only decarbonises at 3%/year**? Then the gap between the CTB/PAB index and the regular market index keeps growing every year.

The consequence: To maintain its trajectory, the CTB/PAB index must exclude more and more companies, deviate further and further from the market. Its Active Share and Tracking Error both increase over time as the market fails to decarbonise at the required pace.

"The Active Share must be the variable" — it is the quantity that adjusts to absorb the widening gap between where the index needs to be (−7%/year) and where the market actually is. The more the market lags, the more different the CTB/PAB index must become from its parent.

LYXOR ETF'S THREE ESG PILLARS (PRACTICAL EXAMPLE)

How a real ETF provider organises its ESG product range — three rows for three different investor goals:

Broad exposures
"Make a widespread impact." Mainstream ESG across the whole portfolio. Products: MSCI ESG Leaders (equity), ESG-screened credit (fixed income), ESG Low TE (tracks market closely with ESG filter). For investors who want broad ESG without big bets.

Climate-focused
"Help limit global warming." Specifically targeting climate. Products: Climate Transition ETF (equity), Green Bonds ETF (fixed income). Maps to CTB/PAB framework. For investors with a specific climate mandate.

Sustainable themes
"Make a targeted impact." Single-theme ETFs targeting specific problems: water sector, clean energy, gender equality, future mobility, smart cities. For investors who want to direct capital at one specific sustainability issue.

Key lesson: ESG is not one product — it is a full spectrum from broad tilts to laser-focused thematic bets. A complete ETF provider must cover the entire range because different institutional clients have different sustainability mandates.

MCQ — SECTION 08B · 5 QUESTIONS

EU ESG Framework & Climate Methods

Q1. Under MiFID II, what are financial advisors now required to ask clients before recommending products?

- A Their net worth and income
- B Their ESG and sustainability preferences
- C Their view on interest rates
- D Their preferred asset class

Q2. What is the key difference between CTB and PAB regarding fossil fuels?

- A CTB excludes fossil fuels; PAB does not
- B CTB has no fossil fuel exclusions; PAB strictly excludes coal, oil, high-carbon electricity
- C Both exclude all fossil fuels equally
- D Neither excludes fossil fuels

Q3. The "Drive the Transition" approach (MSCI Climate Action Index) includes what share of companies?

- A Top 25% ESG performers per sector
- B Top 50% best prepared for the climate transition per sector
- C All companies with a 30% carbon reduction
- D Only renewable energy companies

Q4. What happens to a CTB/PAB index's Active Share if the market decarbonises slower than 7%/year?

- A Active Share decreases — the index converges toward the market
- B Active Share increases — the index must deviate more and more from the market
- C Active Share stays fixed — it is set at launch
- D The index is dissolved and relaunched

Q5. In Lyxor's ESG pillars, which category targets specific themes like water, clean energy, or gender equality?

- A Broad exposures
- B Climate-focused solutions
- C Sustainable themes
- D ESG Leaders

SECTION 09

ETF Mechanics: Replication

PHYSICAL — FULL REPLICATION

→ Best for small indices (<100 securities)
→ Buys every security in the index
→ Securities lending → extra income (positive TD impact)
→ Collateral must be liquid (e.g. Japanese govies)

PHYSICAL — SAMPLING

→ For large indices — skip illiquid/taxed stocks
→ Avoid French FTT (40bps per buy)
→ Avoid illiquid stocks with wide spreads
→ Can skip 15% of index; still achieve low TE

SYNTHETIC — FUNDED ⚠

→ PM gives cash to investment bank counterparty
→ Bank promises to deliver index return at future date
→ Counterparty risk potentially 100%
→ Collateral held at custodian, not owned by PM
→ EMIR regulation manages this risk
→ Mostly replaced by unfunded model

SYNTHETIC — UNFUNDED ✅ SAFER

→ PM passes cash AND buys substitute basket
→ ETF owns the substitute basket outright
→ Total Return Swap has 2 legs; reset daily
→ Valuation gap capped at 10% (UCITS)
→ Now the dominant synthetic model

PEA synthetic trick: MSCI World = 60% US (not EU) → can't go in PEA normally. Synthetic ETF uses European equities as substitute basket → qualifies for PEA while giving MSCI World exposure.

Key equivalence: Physical fund with 100% securities lending ≈ functionally identical to a synthetic fund. Both carry counterparty risk — UCITS caps it at 10%.

MCQ — SECTION 09 · 5 QUESTIONS

ETF Mechanics: Replication

Q1. In an unfunded synthetic ETF, what does the ETF portfolio actually hold?

- A The exact securities of the target index
- B Only cash held at the counterparty bank
- C A substitute basket of securities owned outright by the ETF
- D Futures contracts on the index

Q2. Why does sampling replication sometimes avoid French equities?

- A French companies have poor ESG ratings
- B France has a financial transaction tax of 40bps per purchase
- C French equities are not UCITS-eligible
- D The French market is too illiquid

Q3. What is the UCITS maximum counterparty risk limit for derivative exposure?

- A 5%
- B 10%
- C 20%
- D 100%

Q4. How can an MSCI World ETF qualify for a French PEA?

- A It cannot — PEA requires 100% European stocks
- B Synthetic replication: substitute basket = European equities, satisfying the 75% rule
- C The AMF grants a waiver for large ETFs
- D By hedging US exposure with European put options

Q5. Which synthetic model is considered safer, and why?

- A Funded — the bank holds all the cash
- B Unfunded — ETF owns the substitute basket outright, limiting default exposure
- C Both equally safe under EMIR
- D Physical is always safer

SECTION 10

Primary & Secondary **Markets**

Arbitrage mechanism: If ETF trades above NAV (premium), an AP shorts the ETF and buys the underlying → creates new ETF shares to deliver → flattens position. This keeps ETF price close to NAV at all times.

AP

Authorised Participant — ONLY market participant able to create or redeem ETF shares. JPMorgan = biggest AP globally.

Creation in Cash

AP delivers cash → ETF issuer creates new shares. Simple but involves trading costs.

Creation in Kind ⭑

AP delivers a basket of underlying securities → receives ETF shares. No cash trading costs. Market maker uses their hedge as payment. Key ETF cost advantage. PM can accept a custom/proxy basket if it matches index risk.

COVID example: Underlying high-yield bonds stopped trading — HY ETFs kept trading. Market makers priced risk with wider spreads. Custom creation baskets provided liquidity when underlying markets were frozen. Jane Street (world's biggest market maker) dominated.

5 WAYS TO TRADE A UCITS ETF

1. **Order Book** — Standard exchange, limit/market orders, like a stock.
2. **Intraday at Risk** — Immediate price from broker. Hard to compare (price moves instantly).
3. **NAV Trading** — Price referenced to next closing NAV ± spread. Easy to compare. No intraday risk.
4. **Agency OTC** — Direct OTC. Risk: broker default = trade may not exist.
5. **Algo Trading** — Large orders (e.g. €10M) executed gradually to reduce market impact.

SETTLEMENT INFRASTRUCTURE

CCP — CENTRAL COUNTERPARTY CLEARING

→ Reduces counterparty risk
→ Nets out offsetting trades
→ Default waterfall: initial margin → variation margin → default fund → CCP capital
→ Very solid system

CSD — CENTRAL SECURITIES DEPOSITORY

→ "Back office" of the stock exchange
→ All ETF shares registered here
→ Brokers and custodians hold accounts here
→ Blockchain could replace this in future

Spread drivers: Bid-ask of underlying + creation fees + settlement costs + inventory + taxes (UK stamp duty 50bps, French FTT 40bps) + FX + market maker margin. More market makers = tighter spread (liquidity attracts liquidity).

MCQ — SECTION 10 · 5 QUESTIONS

Primary & Secondary Markets

Q1. Who are the ONLY market participants allowed to create or redeem ETF shares?

- A Market makers
- B The ETF issuer itself
- C Authorised Participants (APs)
- D Any institutional investor

Q2. What is the main advantage of Creation in Kind vs Creation in Cash?

- A Faster to settle
- B AP delivers underlying securities — no trading costs; market maker uses hedge as payment
- C Avoids UCITS regulation
- D Allows shares to be created at a discount to NAV

Q3. What keeps an ETF's market price close to its NAV?

- A Regulatory price controls by the AMF
- B The AP creation/redemption arbitrage mechanism
- C The ETF issuer buys back shares when price falls
- D Daily NAV announcements force a price reset

Q4. What are the two roles of the CCP?

- A Custody of shares and issuing new ETF units
- B Reducing counterparty risk AND calculating net balances from transactions
- C Publishing iNAV and PCF
- D Regulating market makers and enforcing UCITS

Q5. During COVID-19, what demonstrated a key advantage of ETFs over holding underlying bonds directly?

- A ETF fees were temporarily waived
- B Central banks directly purchased ETFs
- C High-yield bond ETFs kept trading even as the underlying bonds stopped — providing liquidity
- D ETFs were converted into physical gold

SECTION 11

BCG Report: From Recovery to Reinvention

Source: BCG Global Asset Management Report 2025, 23rd Edition (April 2025)

THE YEAR IN REVIEW — 2024

$128T
GLOBAL AUM RECORD (2024)

+12%
AUM GROWTH IN 2024

>70%
REVENUE GROWTH FROM MARKETS

+22%
PROFIT GROWTH IN 2024

Growth drivers: S&P 500 +23%, NASDAQ +29% in 2024. Of the $58bn revenue increase, over 70% came from market performance, only 30% from net flows. Half of that gain was offset by fee compression and shift to lower-priced products.

Active exodus: Active AuM declined from 65% to 61% of mutual fund + ETF market in 2024. Net flows: $0.1T outflows from active (ex-MMF) vs $1.6T inflows to passive. North America alone saw $337bn in active outflows — enough to drag global net flows negative.

Fee pressure: Fees on 2024 net inflows were on average 40bps lower than fees on existing 2023 AuM. Revenue per basis point of AuM: fell from 31bps (2010) to 24bps (2024). Structural not cyclical pressure.

THREE FORCES RESHAPING THE INDUSTRY

1 **Product & Distribution Shift**
Investors demanding low-cost products like ETFs. The economics of the industry tilt toward the entity that owns the investor relationship — the distributor. Two big opportunities: (a) active ETFs, model portfolios, separately managed accounts; (b) private assets for retail clients.

2 **Industry Consolidation**
M&A driven by 5 objectives: broaden alternative offerings, expand global presence, build tech/data capabilities, secure permanent capital (insurer partnerships), enhance proximity to clients. Average asset manager doubled AuM from 2013–2023.

3 **Radically Leaner Cost Structures**
Total costs growing at 6% CAGR (2022–2024). Three zero-based levers: outsourcing non-core functions, automation via GenAI, avoiding dual-run costs. Three strategic archetypes emerging below.

THE ACTIVE ETF OPPORTUNITY

Active ETF facts: AuM grew at 39% CAGR over 10 years. No structural difference in performance potential vs active mutual funds — but fees are 0.64% vs 1.08% for mutual funds (−41%). 44% of all ETFs launched in 2024 were actively managed. Yet only 7% of total ETF AuM is active — still nascent and fragmented (top 10 control 65%).

The dilemma: Launching active ETFs means cannibalising existing higher-fee mutual fund business. But from a long-term perspective, active ETFs capture otherwise unavailable capital flows and build closer client relationships.

THREE COST STRUCTURE ARCHETYPES

ARCHETYPE	FOCUS	KEY COST (% OF TOTAL)	STRATEGY
Alpha Shops	IM&TE (investment mgmt & trade execution)	~39%	Generate alpha; use distribution powerhouses for distribution
Beta Factories	IT (scalability & automation)	~22%	Drive down costs; become tech providers for industry

Distribution Powerhouses	Sales & marketing + operations	~39%	Own client relationships; gateway between alpha and beta

SCALE & CONSOLIDATION DYNAMICS

Scale curve: Firms below $300bn: scale reduces costs. $300bn–$500bn: "diseconomies" of scale (rising complexity). Above $500bn: optimisation at scale returns. Largest firms carry higher IT costs (proportionally) — need advanced infrastructure.

PRIVATE ASSETS FOR RETAIL — THE NEXT FRONTIER

Alternative assets generate **more than 50% of global revenues** but hold less than 25% of AuM. Private equity, private debt, real estate, and infrastructure grew at 11.1% CAGR (2014–2024) vs 6.5% for the rest of the industry. 48% of global assets are controlled by retail wallets — mostly untapped for private markets.

CURRENT STRUCTURES

→ Technology-enabled feeder funds (pool assets, no interim liquidity)

→ Evergreen semi-liquid funds (grew 5× in 4 years to $300bn NAV in 2024)

FUTURE STRUCTURES

→ Private assets in active ETFs or model portfolios

→ Hybrid public-private evergreen vehicles

→ Tokenised funds and fractional private assets

→ Automated private credit pools

BCG FORECAST TO 2029

Expected leaders (AuM growth 2024–2029): Equity ETFs and Fixed Income ETFs are projected among the fastest-growing categories, alongside private debt, infrastructure, and private equity — which also carry the highest revenue margins (100–150bps vs <10bps for passive ETFs).

MCQ — SECTION 11 · 5 QUESTIONS

BCG Report: From Recovery to Reinvention

Q1. According to BCG, what share of 2024 revenue growth was driven by market performance (not net flows)?

- A 30%
- B 50%
- C Over 70%
- D 90%

Q2. What was the average fee difference between new 2024 inflows and existing 2023 AuM?

- A New flows paid 40bps less on average
- B New flows paid 40bps more on average
- C No difference — fees were stable
- D New flows paid 10bps more

Q3. Active ETF fees are on average 0.64% vs 1.08% for active mutual funds. How much cheaper is that?

- A About 10% cheaper
- B About 20% cheaper
- C About 41% cheaper
- D About 60% cheaper

Q4. Which BCG cost archetype focuses primarily on IT and aims to become a tech provider for the industry?

- A Alpha Shop
- B Beta Factory
- C Distribution Powerhouse
- D Solutions Provider

Q5. According to BCG, at what AuM threshold do "diseconomies of scale" emerge before efficiency returns above $500bn?

- A $100bn
- B $300bn
- C $500bn
- D $1 trillion

SECTION 12

Larry Fink: 2025 & 2026 Letters

Larry Fink is CEO and founder of BlackRock. His annual letters are read by every finance professional in the world — referenced directly in the course.

2025 LETTER — "THE DEMOCRATIZATION OF INVESTING"

Central thesis: Capital markets have historically worked, but for too few people. Fink's mission is to expand access to investing — especially to private markets — to allow more people to participate in economic growth. "The real payoff is only just beginning."

KEY THEMES: 2025 LETTER

1. **Private Markets Are the Next Frontier**
   Only a tiny fraction of companies are publicly traded — and that fraction is shrinking. 81% of US companies with >$100M revenue are privately held (even higher in EU and UK). Private credit projected to more than double by end of decade. Yet most retail investors cannot access these returns.

2. **New Portfolio Model: 50/30/20**
   Fink argues the classic 60/40 (stocks/bonds) portfolio may no longer represent true diversification. He proposes 50% stocks, 30% bonds, 20% private assets (real estate, infrastructure, private credit). Private assets added 0.5% annual return over 40 years = 14.5% more in your 401(k) = 9 extra years of retirement funded.

3. **Tokenisation = Revolutionising Investing**
   "Every stock, every bond, every fund — every asset — can be tokenised. If they are, it will revolutionise investing." Tokenisation enables fractional ownership, instant settlement, reduced friction. BlackRock launched the first Bitcoin spot ETF in 2024. Blockchain-based systems far more efficient than legacy infrastructure like SWIFT.

4. **Dollar Dominance Under Threat**
   "If the US doesn't get its debt under control, if deficits keep ballooning, America risks losing that position [reserve currency] to digital assets like Bitcoin." A warning embedded in a letter otherwise avoiding explicit politics.

5. **BlackRock's Strategic Pivot**
   BlackRock is no longer only a traditional asset manager. In 14 months: acquired GIP (infrastructure), HPS (private credit), Preqin (private market data). Alternatives to become a top-5 provider globally with $600bn in client assets. $25 trillion is sitting idle in US banks and money market funds — an opportunity.

> "*The reason for the exclusivity has always been risk. Illiquidity. Complexity. But nothing in finance is immutable. Private markets don't have to be as risky. Or opaque. Or out of reach.*"
>
> — Larry Fink, 2025 Annual Letter

2026 LETTER — "GROWING WITH YOUR COUNTRY: THOUGHTS FROM A LONG-TERM OPTIMIST"

Central thesis: In a world of anxiety, constant short-term noise, and geopolitical disruption — the answer is long-term investing. Countries that link citizen wealth to national growth create a "civic miracle." Fink calls for expanding market participation so more people can grow with their economies.

KEY THEMES: 2026 LETTER

1. **Short-termism Is the Enemy**
   "Speed can distort perspective, crowding out long-term thinking." Daily market moves are treated as signals of lasting change. The 10 best days in the S&P 500 over 20 years: miss them and you earn less than half the return. Staying invested matters far more than timing.

2. **Wealth Inequality as the Core Problem**
   Since 1989, $1 in the US stock market has grown more than 15× the value of $1 tied to median wages. AI threatens to repeat this pattern at even larger scale — concentrating wealth among companies and investors positioned to capture it. Capitalism is working — just not for enough people.

3. **Long-Term Investing as a Civic Miracle**
   When people invest their savings over decades, capital markets put that money to work — financing companies, infrastructure, jobs. When this happens in your own country, your future and your nation's future become linked. Fink cites his own parents as an example: modest savings invested through the 1950s–60s US boom = comfortable retirement through compounding.

4 **Barriers to Participation Must Fall**

~40% of the US population has zero exposure to capital markets. Around the world, participation is far lower. People saving in bank accounts earning little, rather than investing. Solutions: emergency savings accounts, investment accounts seeded at birth, digital wallets enabling investing as easily as sending a payment.

5 **Tokenisation + Digital Infrastructure**

Half the world carries a digital wallet. Fink envisions the same wallet enabling broad portfolio investment. Tokenisation updates the "plumbing" of the financial system: investments easier to issue, trade, and access. This connects directly to BlackRock's strategic bet on private market democratisation.

6 **The Old Globalisation Is Fracturing**

Countries spending enormous sums to become self-reliant in energy, defence, and technology. This reordering of global trade has been building for decades — it's not just today's headlines. Long-term investors must look through this noise.

> "*At its best, long-term investing performs a kind of civic miracle. When people invest their savings — over decades, not days — the capital markets put that money to work, financing companies, infrastructure, and jobs. And when that cycle happens in your own country, your future and your nation's future become linked.*"
>
> — Larry Fink, 2026 Annual Letter

CONNECTING BOTH LETTERS TO THE COURSE

LETTER THEME	LINK TO ETF LECTURE
Passive investing dominance	SPIVA data, Vanguard Effect, fee compression
Active ETFs as growth opportunity	BCG report, 1 in 3 new ETFs is active in US
50/30/20 portfolio model	Alternatives as 58% of profitability; BCG growth forecasts
Democratisation / Retail wave	French PEA, 10M European monthly savers, rebate ban
Tokenisation of assets	Blockchain replacing CCP/CSD; BlackRock Bitcoin ETF
Private markets for retail	BCG: semi-liquid funds, feeder funds, active ETF structures
Long-term investing civic purpose	ETFs as democratising tool — cheap, accessible, diversified

MCQ — SECTION 12 · 5 QUESTIONS

Larry Fink Letters 2025 & 2026

Q1. What portfolio model does Fink propose in his 2025 letter to replace the traditional 60/40?

- A 70/30 (stocks/bonds)
- B 50/30/20 (stocks/bonds/private assets)
- C 40/40/20 (stocks/bonds/cash)
- D 60/20/20 (stocks/bonds/alternatives)

Q2. According to Fink's 2025 letter, what % of US companies with over $100M in revenue are privately held?

- A 41%
- B 61%
- C 81%
- D 91%

Q3. In his 2026 letter, Fink states that missing just the 10 best days of the S&P 500 over 20 years results in:

- A A slightly lower but similar return
- B Earning less than half the total return
- C A 30% reduction in final wealth
- D Breaking even with no gains

Q4. According to Fink's 2026 letter, since 1989 how much has $1 in the US stock market grown relative to $1 tied to median wages?

- A 3× more
- B 5× more
- C 15× more
- D Same rate

Q5. What warning does Fink make about the US dollar in his 2025 letter?

- A The dollar will be replaced by the euro
- B If US debt keeps rising, America risks losing reserve currency status to digital assets like Bitcoin
- C The Federal Reserve should abolish the dollar
- D The dollar is overvalued by 20%

QUICK REFERENCE

Key Numbers, Formulas & Concepts

METRIC	VALUE	CONTEXT
Total AuM (global)	$128T	Record 2024; was $36T 20y ago
ETF total market	$20T	≈20% of all AuM; $10T in US
SPIVA US (15y)	10% beat index	Europe 10y = only 3%
Passives profit share	5%	But 23% of AuM
Alternatives profit share	58%	But only 18% of AuM
UCITS max counterparty risk	10%	Derivative/synthetic ETFs
UCITS 5/10/40 rule	Max 10% / <40% sum	Diversification requirement
AMF max Tracking Error	<1% (5% volatile)	To qualify as index fund
SPY spread	0.5bps	1 cent on $500 price
EU CTB day-1 cut	−30% carbon	+ −7%/year trajectory
EU PAB day-1 cut	−50% carbon	+ −7%/year trajectory
French PEA limit	€150,000	75% EU assets required; 5y hold
iShares EU market share	42%	Top 3 = 65% of EU market
EU ETF net flows (latest)	$333bn	vs $259bn previous year
Active fund avg fee	71.4bps	vs passive 13.5bps (5× more)
BGI acquisition	$13.5bn (2010)	BlackRock → #1 ETF firm
Vanguard avg TER	<10bps	Structural cost advantage
Active ETF fee (BCG)	0.64%	vs 1.08% for active mutual funds (−41%)
Fink: 50/30/20 portfolio	Stocks/Bonds/Private	New diversification standard
Fink: idle US capital	~$25 trillion	Sitting in banks & MMFs
Global infra demand by 2040	$68 trillion	Fink 2025 letter
Stocks vs wages since 1989	15× more growth	Fink 2026 letter — inequality thesis

KEY FORMULAS

Active Share

Σ |portfolio weight − index weight| ÷ 2. Range: 0% (= index) → 100% (completely different from index).

Tracking Difference

ETF return − Index return. Should be ≈ 0. Hurt by: TER. Helped by: securities lending income.

Tracking Error

Annualised standard deviation of (ETF daily return − Index daily return). Measures consistency of replication over time.

SPY spread cost

Spread in cents ÷ Price in $ = % round-trip cost. Example: 1 cent ÷ $500 = 0.002% = 0.2bps per side = 0.5bps total.

ANSWER KEY

MCQ Answers & Explanations

All 60 answers. Check your work section by section.

SECTION 01 — THE ASSET MANAGEMENT INDUSTRY

Q1. Global AuM size

C $128 trillion

Grew from $36T twenty years ago. 7%/year growth mostly from market appreciation not flows (1–3% net).

Q2. Passive profitability share

B 5%

Passives hold 23% of AuM but generate only 5% of profit. Low fees = low margin despite massive scale.

Q3. Alternatives profitability

D 58%

18% AuM but 58% of all profits. Private equity is the most profitable business in asset management.

Q4. Passive + alts combined share

C 41%

Was 20%. Now 41%. Active Core squeezed from both extremes simultaneously.

Q5. Net flows into passives

C 70–80%

Between 70–80% of all new money flows into passive products, creating relentless fee compression.

SECTION 02 — WHY PASSIVE INVESTING IS WINNING

Q1. EU active beat index (10y)

D 3%

SPIVA: only 3% of European active managers beat index over 10 years. US figure = 10% over 15 years.

Q2. Active Share = index

A 0%

Σ|weight − index weight| ÷ 2. If all weights are identical → all differences = 0 → Active Share = 0%.

Q3. Fixed income post-COVID

B Rising rates = more differentiation

Near-zero rates made all bonds look alike. Sharp rate rises created skill differentiation opportunities.

Q4. Best market for alpha

C Indian equities

~50% of active managers beat Indian index. Less coverage = information asymmetry = alpha. Same for small caps, EM, Switzerland.

Q5. Closet index fund

C Too large to deviate, charges active fees

Fund so big its Active Share → 0%. Effectively tracks index — no justification for active management fees.

SECTION 03 — KEY PLAYERS: BLACKROCK & VANGUARD

Q1. How BlackRock became #1 ETF

C Acquired BGI (iShares) $13.5bn 2010

BGI invented iShares. Described as "the deal of the century" in the lecture.

Q2. Why Vanguard is private

B Fiduciary — serve only one master

"No man can serve two masters." Listed firms serve shareholders AND clients. Vanguard mutual structure: one interest only.

Q3. Bogle's insight

B Most mutual funds don't beat index after fees

Bogle's study showed active underperform after fees. Vanguard 500 launched 1975 — raised only $11M vs $150M target.

Q4. Two growth strategies

C ETFs and Alternatives

These are the two areas of genuine growth. All major players (Amundi, Deutsche Bank, etc.) are developing both.

SECTION 04 — WHAT IS AN ETF?

Q1. TD vs TE

B TD = performance gap; TE = volatility of that gap

TD is pure performance (ETF − index return). TE is annualised std dev of TD — measures consistency, not average gap.

Q2. AMF max Tracking Error

B 1% (5% for volatile indices)

AMF sets 1% for standard indices, 5% for highly volatile ones, to ensure labelled index funds actually track their index.

Q3. 5/10/40 rule limit

D 40%

No single position above 10%. All positions 5–10% must sum to less than 40%. Ensures diversification.

Q4. PCF – what and when

B Portfolio Composition File, before 9am daily

ETF issuers must publish PCF before 9am. Market makers use it to know exactly what's in the ETF for intraday pricing.

Q5. Positive impact on TD

B Securities lending income

Lending fee income can offset or exceed the TER, resulting in a positive TD (ETF outperforms index net of fees).

SECTION 05 — ETF GROWTH & MARKET STRUCTURE

Q1. UCITS tax advantage

B Tax-protected until sale

US funds are tax-transparent — gains taxed annually. UCITS funds are protected until you sell. Key reason Asian investors use UCITS ETFs.

Q2. Wealth Wave mechanism

C Rebate ban forces cheapest recommendation

Banks receive 1–2%/year kickback from active managers (vs 25bps for ETFs). When banned, advisors choose cheapest products → ETFs.

Q3. iShares EU market share

C 42%

iShares holds 42% of European ETF market. Top 3 of 107 providers = 65% of market — extremely concentrated.

Q4. EU ETF net flows (latest)

C $333bn

€333bn collected in the most recent year, up from $259bn the prior year — ~28% growth in annual inflows.

SECTION 06 — RETAIL EVOLUTION & PEA

Q1. PEA holding period

C 5 years

After 5 years: only social tax (~15.5%) instead of 30% flat tax — roughly halves the tax burden on gains.

Q2. MSCI World in PEA

B Synthetic replication, substitute basket = EU equities

The actual portfolio holds European equities (satisfying 75% EU rule). A swap provides MSCI World performance.

Q3. PEA maximum

C €150,000

PEA capped at €150k total investment. PEA-PME is a separate sibling account for small-cap companies.

Q4. French investors under 35

C 45%

Almost 45% of active French ETF investors are under 35, driven by pension concerns and easy mobile access. Avg 4 transactions/year.

SECTION 07 — THE INDEX INDUSTRY

Q1. Stock 100, pays 3€ dividend – PR index

B **97**

Price Return index: dividends not reinvested. Stock price drops by dividend on ex-dividend date: 100 − 3 = 97.

Q2. Index using Total Return

C **DAX**

DAX is unusually a Gross Total Return index. Makes DAX performance look significantly better than CAC 40 (PR) over time.

Q3. Why NTR is easiest to beat

B **Applies maximum withholding tax on dividends**

NTR uses the highest possible withholding tax rate. Most investors pay a lower rate — so they naturally "beat" this benchmark.

Q4. Nikkei weighting method

C **Price weighted**

Nikkei is price-weighted — higher share price = larger weight regardless of company size. Historical convention, considered flawed.

Q5. China under-represented in MSCI

C **EM status, non-convertible currency, restricted access**

China's GDP ≈ US but MSCI classifies it as emerging and applies a large discount due to currency controls and limited foreign access.

SECTION 08 — ESG & CLIMATE INDICES

Q1. EU PAB day-1 carbon cut

D **−50%**

PAB = −50% day one. CTB = −30% day one. Both require −7%/year thereafter.

Q2. Annual trajectory for CTB and PAB

C **−7%/year**

Both require −7%/year. Derived from Paris Agreement: net zero by 2050, 45% cut by 2030 from 2010 baseline.

Q3. Scope 3 definition

C **Indirect value chain emissions**

Scope 1 = direct (own ops), Scope 2 = purchased energy, Scope 3 = entire upstream + downstream value chain. Largest and hardest to measure.

Q4. Top 25% ESG per sector

D **SRI**

SRI is the most restrictive — top 25% per sector, excludes ~75% of universe. ESG Leaders = top 50%. ESG Screened = minimal exclusions.

SECTION 09 — ETF MECHANICS: REPLICATION

Q1. Unfunded synthetic ETF holds

C **Substitute basket, owned outright**

ETF buys a substitute basket of real securities. Owns these outright. A swap exchanges basket return for index return. ETF owns real assets.

Q2. Sampling avoids French equities

B **French FTT = 40bps per purchase**

40bps tax per French equity purchase. By excluding some, ETF avoids this cost → improves TD without meaningfully increasing TE.

Q3. UCITS max counterparty risk

B **10%**

UCITS limits derivative counterparty exposure to 10% of NAV. Swap in synthetic ETF reset daily to ensure gap never exceeds 10%.

Q4. MSCI World in PEA

B **Synthetic, substitute basket = EU equities**

Portfolio holds European equities (satisfying 75% PEA rule). Swap exchanges European basket return for MSCI World return.

Q5. Safer synthetic model

B **Unfunded — ETF owns substitute basket outright**

Funded = collateral pledged but not owned. If bank defaults, recovery uncertain. Unfunded = ETF owns basket, no asset default risk.

SECTION 10 — PRIMARY & SECONDARY MARKETS

Q1. Who can create/redeem ETF shares

C **Authorised Participants (APs)**

APs are the gatekeepers of ETF share creation and redemption. JPMorgan = largest AP. This exclusive role makes the arbitrage mechanism work.

Q2. Creation in Kind advantage

B AP delivers securities — no trading costs; hedge as payment

No cash trading costs. Market maker delivers their existing hedge as payment. Everyone saves costs. Even more powerful for fixed income.

Q3. Keeps ETF price close to NAV

B AP creation/redemption arbitrage

If ETF trades above NAV, AP shorts ETF + buys underlying → creates shares to deliver → price normalises. Automatic and continuous.

Q4. Two CCP roles

B Reduces counterparty risk + nets transactions

CCP insures trades and nets out offsetting positions, dramatically reducing the amount of securities and cash that need to change hands.

Q5. COVID ETF advantage

C HY bond ETFs kept trading when underlying stopped

ETFs provided liquidity precisely when underlying markets failed. Custom creation baskets allowed market makers to price risk even without perfect hedges.

SECTION 11 — BCG REPORT: FROM RECOVERY TO REINVENTION

Q1. Share of revenue from market performance

C Over 70%

BCG: over 70% of $58bn revenue increase came from market performance (S&P +23%, NASDAQ +29%). Only 30% from net inflows.

Q2. New vs existing AuM fee gap

A New flows paid 40bps less on average

Fees on 2024 net inflows averaged 40bps lower than fees on existing 2023 AuM — a clear structural revenue pressure signal.

Q3. Active ETF vs mutual fund fee difference

C About 41% cheaper

(1.08 − 0.64) / 1.08 = 40.7% ≈ 41%. Same performance potential, significantly lower fees — a strong value proposition.

Q4. IT-focused archetype

B Beta Factory

Beta Factories invest heavily in IT (~22% of costs) to drive scalability and automation. Can become tech providers for the broader industry.

Q5. Diseconomies of scale threshold

B $300bn

Firms below $300bn: scale reduces costs. $300bn–$500bn: diseconomies emerge (rising complexity). Above $500bn: optimisation at scale returns.

SECTION 12 — LARRY FINK LETTERS 2025 & 2026

Q1. Fink's 2025 portfolio model

B 50/30/20 (stocks/bonds/private)

Classic 60/40 may no longer represent true diversification. 20% in private assets (real estate, infrastructure, private credit) adds ~0.5%/year = 9 extra retirement years over 40 years.

Q2. US private companies (2025)

C 81%

81% of US companies with over $100M in revenue are privately held. Even higher in EU and UK. Most companies now grow via private markets, not IPOs.

Q3. Missing 10 best S&P 500 days (2026)

B Less than half the total return

Over 20 years, every $1 grew 8×+ in the S&P 500. Miss just the 10 best days: earn less than half. Some best days came amid the worst headlines.

Q4. Stocks vs wages since 1989

C 15× more

$1 in the US stock market since 1989 grew more than 15× the value of $1 tied to median wages. Core of Fink's inequality thesis in 2026 letter.

Q5. Dollar dominance warning (2025)

B Rising US debt could shift reserve status to Bitcoin

Fink warns: "If the US doesn't get its debt under control, America risks losing that position to digital assets like Bitcoin." Notably cautious for a CEO whose firm launched the first Bitcoin ETF.

SECTION 08B — EU ESG FRAMEWORK & CLIMATE METHODS

Q1. MiFID II advisor requirement

B ESG and sustainability preferences

MiFID II now mandates advisors ask about sustainability preferences before any product recommendation — one of the EU's key levers for redirecting investment flows toward ESG.

Q2. CTB vs PAB on fossil fuels

B CTB has no fossil exclusions; PAB excludes coal, oil, high-carbon electricity

Most important practical difference. CTB only reduces carbon intensity overall. PAB has hard exclusions: coal (1%+ revenues), oil (10%+), natural gas (50%+), electricity above 100gCO2/kWh.

Q3. Drive the Transition – share of companies

B Top 50% best prepared per sector

Climate Action Index includes the 50% of companies best positioned for the climate transition relative to sector peers — not top 50% ESG scorers overall. Keeps sector diversification intact.

Q4. Active Share when market lags

B Active Share increases

If market decarbonises at 3%/year but index must follow -7%/year, the gap widens every year. The index must exclude more companies and deviate further from parent — Active Share and Tracking Error both rise.

Q5. Lyxor pillar for water, clean energy, gender

C Sustainable themes

Sustainable themes = most targeted pillar, single-theme ETFs. Broad exposures = mainstream ESG across portfolio. Climate-focused = climate transition and green bonds specifically.